

## Phase 1: Analysis Memo

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This analysis evaluates 16 experimental runs comparing baseline vs. improved prompts across two tasks (Paper Triage and Claim-Evidence Extraction). Both models persistently deviated from the required (source\_id, chunk\_id) citation format, revealing that Phase 2 RAG system cannot rely on prompting alone.

### Key Performance Patterns and Failure Modes

#### Overall Score Improvements (Baseline → Improved)

- **Groundedness:** 3.25 → 3.38 (+3.9%%)
- **Citation Correctness:** 1.5 → 2.63 (+75%) ← *largest improvement*
- **Usefulness:** 3.50 → 3.9 (+11%)

|                  |                 | Total Score | Groundedness | Citation   | Usefulness |
|------------------|-----------------|-------------|--------------|------------|------------|
| <b>Task 1</b>    | Baseline Prompt | 7.5         | 3            | 1          | 3.5        |
|                  | Improved Prompt | 10.25       | 3.25         | 3.25       | 3.75       |
| <b>Task 2</b>    | Baseline Prompt | 9           | 3.5          | 2          | 3.5        |
|                  | Improved Prompt | 10          | 3.5          | 2          | 4          |
|                  | <b>Average</b>  | <b>9.2</b>  | <b>3.3</b>   | <b>2.1</b> | <b>3.7</b> |
|                  |                 |             |              |            |            |
| <b>All Tasks</b> | Baseline Prompt | 8.25        | 3.25         | 1.5        | 3.5        |
|                  | Improved Prompt | 10.13       | 3.38         | 2.63       | 3.88       |
|                  | <i>Impact</i>   | 22.73%      | 3.85%        | 75.00%     | 10.71%     |

- The Citation dimension had the lowest overall score (2.1), and showed the most variance during the trial runs. With the improved prompts in the first Paper Triage task, the citation dimension benefited most from the explicit format requirements and grounding constraints. However, in the Claim-Evidence Extraction task (which required exact quotations and numerical claims), the citations did not follow the required format across both prompt approaches.
- Improved prompts delivered research-grade performance, and grounding constraints were essential to prevent fabrication more effectively than relying on model defaults. Claim-Evidence extraction scored highest on “Usefulness” due to exact prompt requirements and constraints.

- Claude Sonnet 4.5 outperformed Gemini 3 Pro models exhibited similar patterns, including citation challenges, suggesting prompt engineering principles generalize across models rather than being model-specific.
- While both models performed similarly in the “Usefulness” dimension, there was still a slight variance across the board: In general, Gemini Pro provided better answers to baseline prompts than Sonnet 4.5, and to my surprise, Sonnet 4.5 performed better than Gemini Pro in the improved prompts. I suspect that the sophisticated and detailed improved prompt instructions were sometimes in conflict with Gemini’s “Pro” internal planning mode, degrading the overall output compared to Sonnet. Additionally, Claude’s format was more user-friendly when copied to a document.

## **Phase 2 Design Recommendations**

- <sup>1</sup>Implement citation format post-processor (don't depend on model compliance)
  - Implement parser to convert (filename, page X) → (source\_id, chunk\_id) automatically
- Add quote verification to validate groundedness
  - Check every claim matches substring in retrieved chunk
  - Flag outputs with <90% quote match rate for human review
  - Implement confidence scoring based on exact vs. paraphrased evidence
- Implement multi-stage retrieval pipeline:
  - Stage 1: Retrieve broad context (multiple chunks)
  - Stage 2: Extract specific evidence from retrieved chunks
  - Stage 3: Verify evidence supports claim before including in answer
- Create concrete chunk\_id examples for prompts
- Revise Evaluation approach
  - Track citation precision, recall, and format compliance as key metrics
  - Set 90%+ thresholds for groundedness and citation recall
  - Use Phase 1 improved prompts as baseline for Phase 2 generation prompts

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<sup>1</sup> Used Claude Sonnet 4.5 for Rag design recommendations.